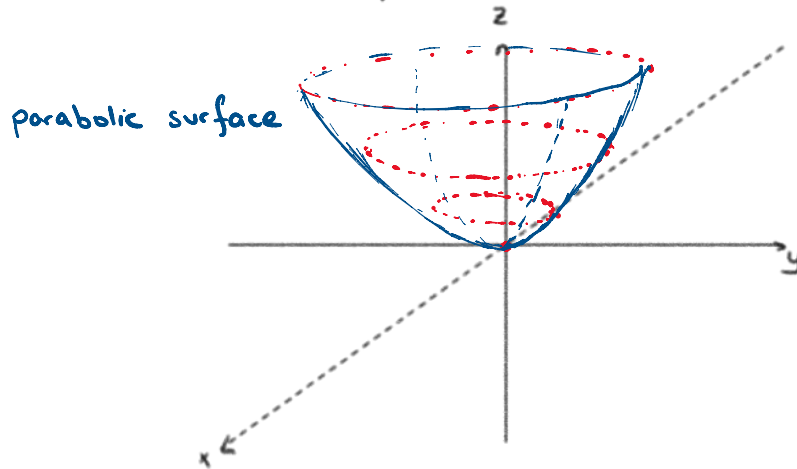


Surface Representation and Surface Integrals

Tuesday, June 3, 2025 4:17 AM

16. Surfaces: being 2D in a 3-dimensional world

16.1 What does $z = x^2 + y^2$ look like?

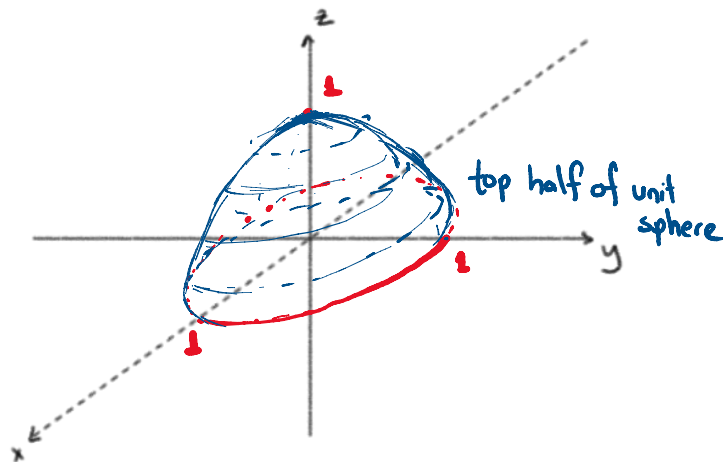


What does $z = \sqrt{1 - x^2 - y^2}$ look like?

Note: $1 - x^2 - y^2 \geq 0 \Rightarrow x^2 + y^2 \leq 1$

and $z^2 = 1 - x^2 - y^2 \Rightarrow x^2 + y^2 + z^2 = 1$ (equation of sphere)

but z is not negative



16.2 What if we want the entire sphere?

Use $x^2 + y^2 + z^2 = 1$ without any restrictions!

Rewrite $x^2 + y^2 + z^2 - 1 = 0$ to make it look like
 $g(x, y, z) = 0$

REWRITE $x+y+z-1=0$ TO MAKE IT LOOK LIKE
 $g(x,y,z)=0$

16.3 Representations

Surfaces are often represented as $z=f(x,y)$

e.g. $z=x^2+y^2$ (parabolic bowl)

e.g. $z=\sqrt{1-x^2-y^2}$ (half-sphere)

e.g. $z=\sqrt{x^2+y^2}$ (top half-cone)

or as $g(x,y,z)=0$ [also $g(x,y,z)=\text{const}$]

e.g. $x^2+y^2+z^2-1=0$ (sphere)

$x^2+y^2-1=0$ (cylinder)

$x+y+z-1=0$ (plane)

$z^2-x^2-y^2=0$ (double cone)

However, we are more interested in parametric representations:

$$\vec{r}(u,v) = x(u,v)\hat{i} + y(u,v)\hat{j} + z(u,v)\hat{k}$$

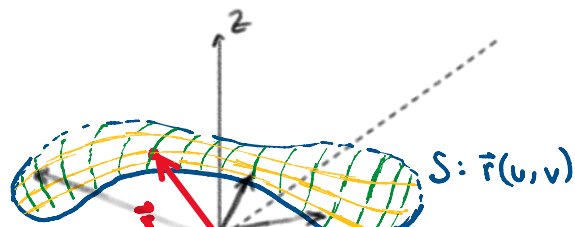
for $(u,v) \in \mathcal{R}$ (subset of $\mathbb{R}^2$)

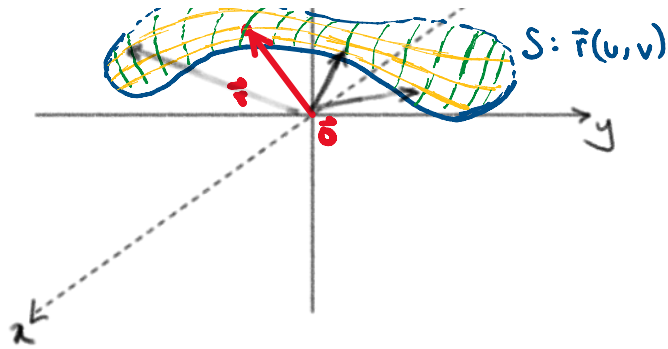
16.4 Typically we would want our parametrizations to look like

$$\vec{r}(u,v) \text{ for } a \leq u \leq b, \quad c \leq v \leq d$$

such that u & v are completely independent

The idea is that $\vec{r}(u,v)$ would try to "trace out" the entire surface by changing u and v





16.5

EXAMPLE 1 Parametric Representation of a Cylinder

The circular cylinder $x^2 + y^2 = a^2$, $-1 \leq z \leq 1$, has radius a , height 2, and the z -axis as axis. A parametric representation is

$$\mathbf{r}(u, v) = [a \cos u, a \sin u, v] = a \cos u \mathbf{i} + a \sin u \mathbf{j} + v \mathbf{k} \quad (\text{Fig. 242}).$$

The components of $\mathbf{r}$ are $x = a \cos u$, $y = a \sin u$, $z = v$. The parameters u, v vary in the rectangle $R: 0 \leq u \leq 2\pi$, $-1 \leq v \leq 1$ in the uv -plane. The curves $u = \text{const}$ are vertical straight lines. The curves $v = \text{const}$ are parallel circles. The point P in Fig. 242 corresponds to $u = \pi/3 = 60^\circ$, $v = 0.7$. ■

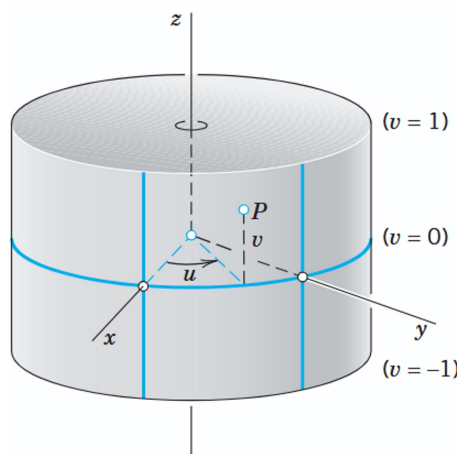


Fig. 242. Parametric representation of a cylinder

EXAMPLE 2 Parametric Representation of a Sphere

A sphere $x^2 + y^2 + z^2 = a^2$ can be represented in the form

$$(3) \quad \mathbf{r}(u, v) = a \cos v \cos u \mathbf{i} + a \cos v \sin u \mathbf{j} + a \sin v \mathbf{k}$$

where the parameters u, v vary in the rectangle R in the uv -plane given by the inequalities $0 \leq u \leq 2\pi$, $-\pi/2 \leq v \leq \pi/2$. The components of $\mathbf{r}$ are

$$x = a \cos v \cos u, \quad y = a \cos v \sin u, \quad z = a \sin v.$$

The curves $u = \text{const}$ and $v = \text{const}$ are the “meridians” and “parallels” on S (see Fig. 243). *This representation is used in geography for measuring the latitude and longitude of points on the globe.*

Another parametric representation of the sphere also used in mathematics is

$$(3^*) \quad \mathbf{r}(u, v) = a \cos u \sin v \mathbf{i} + a \sin u \sin v \mathbf{j} + a \cos v \mathbf{k}$$

where $0 \leq u \leq 2\pi$, $0 \leq v \leq \pi$.

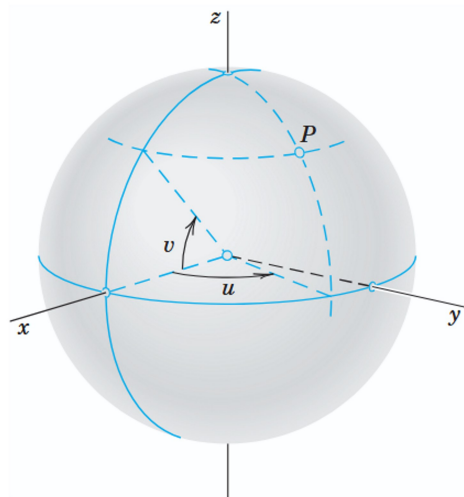


Fig. 243. Parametric representation of a sphere

Slido: Try to create a parametric representation of the
Cone: $z = \sqrt{x^2 + y^2}$, $0 \leq z \leq H$

EXAMPLE 3 Parametric Representation of a Cone

A circular cone $z = \sqrt{x^2 + y^2}$, $0 \leq z \leq H$ can be represented by

$$\mathbf{r}(u, v) = [u \cos v, u \sin v, u] = u \cos v \mathbf{i} + u \sin v \mathbf{j} + u \mathbf{k},$$

in components $x = u \cos v$, $y = u \sin v$, $z = u$. The parameters vary in the rectangle R : $0 \leq u \leq H$, $0 \leq v \leq 2\pi$. Check that $x^2 + y^2 = z^2$, as it should be. What are the curves $u = \text{const}$ and $v = \text{const}$?

17 Tangent Plane and Surface Normal

17.1 The tangent vectors of all the curves on a surface S through a point P of S form a tangent plane of S at P (called $T_P S$)

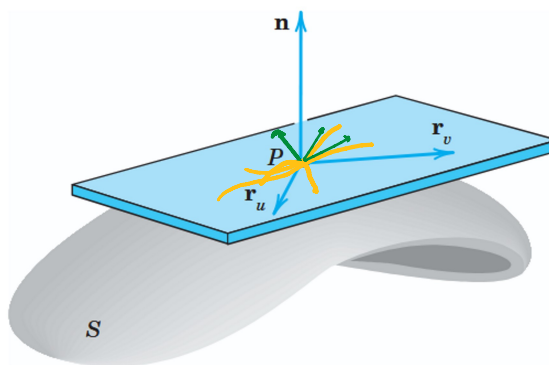


Fig. 244. Tangent plane and normal vector

17.2 If S is parametrized $\vec{r}(u, v)$, then the partial derivatives $\frac{\partial \vec{r}}{\partial u}\bigg|_P$ and $\frac{\partial \vec{r}}{\partial v}\bigg|_P$ (called $\vec{r}_u$ and $\vec{r}_v$) are tangential to S at P . Assuming $\vec{r}_u$ and $\vec{r}_v$ are linearly independent, $\{\vec{r}_u, \vec{r}_v\}$ forms a basis for $T_P S$.

Slide: What do we mean by $\{\vec{r}_u, \vec{r}_v\}$ is a basis for $T_P S$?

17.3 We get a normal vector N of S at P by

$$N = \vec{r}_u \times \vec{r}_v$$

with unit normal vector $\vec{n}$ of S at P :

$$\vec{n} = \frac{1}{|N|} \vec{N} = \frac{\vec{r}_u \times \vec{r}_v}{|\vec{r}_u \times \vec{r}_v|}$$

17.4 Orientation of a Surface (early ideas)

Instead of $\vec{n}$ or $\vec{N}$, we could choose $-\vec{n}$ or $-\vec{N}$ as our choice of normal vector. Our choice of $\vec{n}$ relates to an orientation of S .

In very watered down explanation, choosing an orientation of S involves a consistent way of choosing $\vec{n}$ at each point in S such that $\vec{n}$ varies smoothly

in very weird ways. ...
 consistent way of choosing $\vec{n}$ at each point in S such that $\vec{n}$ varies smoothly over S and at each point $\vec{n}$ is uniquely determined by the choice.

Theory: Nonorientable Surfaces

A sufficiently small piece of a smooth surface is always orientable. This may not hold for entire surfaces. A well-known example is the **Möbius strip**,⁵ shown in Fig. 248. To make a model, take the rectangular paper in Fig. 248, make a half-twist, and join the short sides together so that A goes onto A , and B onto B . At P_0 take a normal vector pointing, say, to the *left*. Displace it along C to the right (in the lower part of the figure) around the strip until you return to P_0 and see that you get a normal vector pointing to the *right*, opposite to the given one. See also Prob. 17.

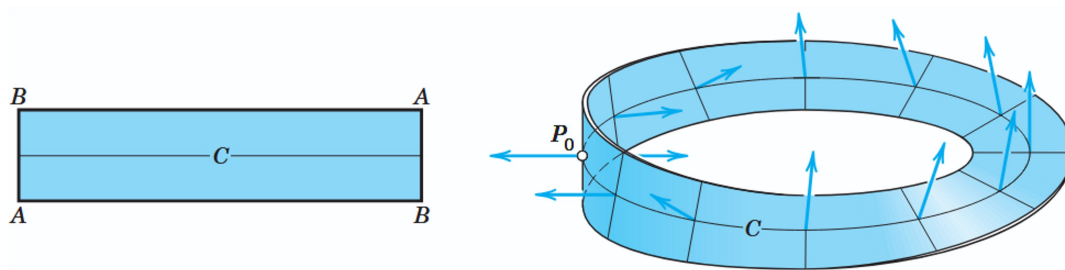


Fig. 248. Möbius strip

17.5

THEOREM 1

Tangent Plane and Surface Normal

If a surface S is given by (2) with continuous $\mathbf{r}_u = \partial \mathbf{r} / \partial u$ and $\mathbf{r}_v = \partial \mathbf{r} / \partial v$ satisfying (4) at every point of S , then S has, at every point P , a unique tangent plane passing through P and spanned by $\mathbf{r}_u$ and $\mathbf{r}_v$, and a unique normal whose direction depends continuously on the points of S . A normal vector is given by (4) and the corresponding unit normal vector by (5). (See Fig. 244.)

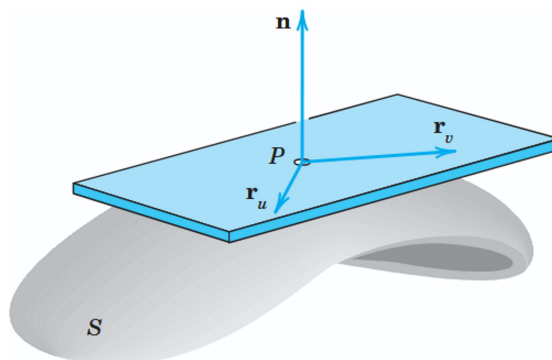


Fig. 244. Tangent plane and normal vector

17.6 Recall that for a surface S represented by $g(x, y, z) = 0$, the gradient $\vec{\nabla}g$ is a vector field that is normal to S .

Thus
$$\vec{n} = \frac{1}{|\vec{\nabla}g|} \vec{\nabla}g.$$

Examples:

EXAMPLE 4 Unit Normal Vector of a Sphere

From (5*) we find that the sphere $g(x, y, z) = x^2 + y^2 + z^2 - a^2 = 0$ has the unit normal vector

$$\mathbf{n}(x, y, z) = \left[\frac{x}{a}, \frac{y}{a}, \frac{z}{a} \right] = \frac{x}{a} \mathbf{i} + \frac{y}{a} \mathbf{j} + \frac{z}{a} \mathbf{k}.$$

We see that $\mathbf{n}$ has the direction of the position vector $[x, y, z]$ of the corresponding point. Is it obvious that this must be the case? ■

EXAMPLE 5 Unit Normal Vector of a Cone

At the apex of the cone $g(x, y, z) = -z + \sqrt{x^2 + y^2} = 0$ in Example 3, the unit normal vector $\mathbf{n}$ becomes undetermined because from (5*) we get

$$\mathbf{n} = \left[\frac{x}{\sqrt{2(x^2 + y^2)}}, \frac{y}{\sqrt{2(x^2 + y^2)}}, \frac{-1}{\sqrt{2}} \right] = \frac{1}{\sqrt{2}} \left(\frac{x}{\sqrt{x^2 + y^2}} \mathbf{i} + \frac{y}{\sqrt{x^2 + y^2}} \mathbf{j} - \mathbf{k} \right). \quad \text{■}$$

We are now ready to discuss surface integrals and their applications, beginning in the next section.

17.7 Surface represented by $z = f(x, y)$, we can get a parametrization

$$\vec{r}(u, v) = [u, v, f(u, v)]$$

 and
$$\vec{N} = [-f_u, -f_v, 1]$$

Stdo: Can you justify this?

Example: $z = \sqrt{x^2 + y^2}$
 so that $f(x, y) = \sqrt{x^2 + y^2}$
 Then $f_x = \frac{x}{\sqrt{x^2 + y^2}}$, $f_y = \frac{y}{\sqrt{x^2 + y^2}}$
 and $N = \left[\frac{-x}{\sqrt{x^2 + y^2}}, \frac{-y}{\sqrt{x^2 + y^2}}, 1 \right]$
 which gives $-\vec{n}$ from the previous example.

which gives $-\vec{n}$ from the previous example.